MEDRECORDS

Development Roadmap

Phased build plan from MVP to national scale

1. Roadmap Philosophy

This roadmap is sequenced around three constraints that govern an EMR/PHR rollout in Nigeria:

- Provider adoption is the binding constraint, not technology. Every phase must end with a credible answer to "why do providers add a record this week."
- Capital efficiency matters. Each phase should produce evidence sufficient to justify the next funding round before exhausting the previous one.
- Trust is built one geography at a time. We do not attempt national rollout in phase one. We win Lagos, then Abuja, then expand.

2. Phases at a Glance

Phase	Window	Theme	What ships
Phase 0	Months 0–3	Foundations	Team, infra, regulatory groundwork, design partnerships
Phase 1	Months 3–9	Pharmacy MVP	Patient app + pharmacy module. First 20 pharmacies in Lagos.
Phase 2	Months 9–15	GP Wave	Provider clinician app + first HMS adapter. 50 active providers.
Phase 3	Months 15–24	Payer & Multi-City	First HMO contract. Expand to Abuja and one other city.
Phase 4	Year 2–3	National Scale	5+ cities, multiple payer contracts, hospital deals, HMS adapter library.
Phase 5	Year 3+	Regional & Platform	ECOWAS expansion, public APIs, research platform, AI overlays.

3. Phase 0 — Foundations (Months 0–3)

Pre-product work. The goal is to be ready to build, not to have built.

3.1 Team

- Hire CTO, lead backend engineer, lead mobile engineer, lead product designer, head of clinical operations (registered MDCN doctor).
- Engage a part-time DPO and external NDPA counsel.
- Identify 3 design-partner pharmacies and 2 design-partner clinics.

3.2 Regulatory & Legal

- Incorporate; register with CAC and FIRS.
- Open conversations with NDPC, FMOH digital health unit, NHIA innovation desk.
- Begin NIMC verification API partnership conversations.

- Draft NDPA-compliant privacy policy and terms of service.

3.3 Architecture & Design

- Finalize cloud architecture (provider selection, in-country residency setup).
- Publish v0 of Nigeria FHIR Implementation Guide (open-source).
- Design system established (component library, accessibility tokens).
- Identity and consent model design reviewed with legal counsel.

3.4 Exit Criteria

- Founding team in place.
- Cloud environment provisioned with in-country residency confirmed.
- NDPA registration filed.
- NIMC partnership term sheet (or credible alternative path) signed.
- Three pharmacy and two clinic design partners committed to beta participation.

4. Phase 1 — Pharmacy MVP (Months 3–9)

Smallest end-to-end product that produces real value at the point of dispensing. Patients can have records, but the primary write path is pharmacies recording dispenses against prescriptions.

4.1 Ship List

- Patient app (Android first, iOS in month 6) with NIN onboarding, record viewing, audit log, consent grants.
- Pharmacy module: prescription verification, dispense recording, drug interaction flagging, allergy display.
- Lightweight provider app for issuing electronic prescriptions (used by the design-partner clinics to feed the pharmacy demo).
- Backend: FHIR-shaped Patient, MedicationRequest, MedicationDispense, AllergyIntolerance, Consent.
- Audit logging end-to-end.
- Admin console: facility and user management, basic reporting.
- USSD shortcode: minimal — verify a prescription code, see last 3 records summary.

4.2 Geographic Scope

Two adjacent LGAs in Lagos with high pharmacy density. Aim for at least 20 active pharmacies (recording dispenses weekly) and 5 design-partner clinics writing prescriptions into the system.

4.3 Exit Criteria

- 20+ pharmacies in active use
- 5,000+ patient accounts created (target conversion rate from pharmacy invites)
- 25,000+ dispense events recorded
- Median time from prescription issued to verified-and-dispensed under 24 hours

- Beta NPS baseline established for both patient and pharmacy users
- Independent security audit completed; no critical findings open
- Seed round closed or in close

5. Phase 2 — GP Wave (Months 9–15)

Build out the provider clinician experience and the integration story. GP clinics in the same Lagos LGAs get the full clinician workstation; we ship the first HMS adapter; specialist modules become real.

5.1 Ship List

- Full clinician app: structured encounter (chief complaint, ICD-10 diagnosis, vitals, plan, notes), prescriptions, lab orders, referrals.
- Offline-first sync engine for the clinician app (queue, conflict resolution).
- Specialist modules: dental, optical, basic chiropractic templates.
- HMS adapter #1 (one of the top Nigerian HMS vendors, chosen during phase 1).
- Lab order/result flow with at least 2 lab partners.
- Break-glass implementation with full compliance review workflow.
- iOS patient app at parity with Android.
- Multilingual patient app: Yoruba added (Hausa and Igbo in late phase 2 / early phase 3).

5.2 Exit Criteria

- 50+ active providers (pharmacies + clinics)
- 20+ clinics actively authoring encounters
- 25,000+ patient accounts
- Cross-provider continuity rate above 25% (patients with records from 2+ providers)
- First HMS adapter live with at least 3 paying customer sites
- First payer contract signed or in advanced negotiation

6. Phase 3 — Payer & Multi-City (Months 15–24)

6.1 Ship List

- Payer-grade analytics dashboard: claims-grade encounter and dispense data, scoped to enrolled members with consent.
- Payer API: scoped FHIR read for HMOs, with member-consent enforcement.
- Hospital tier (50+ bed institutions): SLA, dedicated CSM, on-prem read cache option.
- Abuja launch — replicate Lagos playbook.
- One secondary-city launch (Port Harcourt or Ibadan).
- Second HMS adapter.
- Premium patient tier launched (paid family subscription).
- Provider mobile app (companion to web) for clinicians who use phones at the point of care.

6.2 Exit Criteria

- 150+ active providers across 3 cities
- 100,000+ patient accounts
- First payer live in production with at least 50,000 lives under contract
- ISO 27001 certified
- Series A closed
- Provider gross revenue retention above 90%

7. Phase 4 — National Scale (Year 2–3)

7.1 Themes

- Payer-driven distribution: HMOs and NHIA-accredited schemes mandate MedRecords-compatible records for accredited providers, creating top-down pull.
- HMS adapter library: 5+ adapters covering the bulk of the HMS market.
- Hospital deployment at scale: enterprise sales motion, structured deployment methodology, on-prem cache for large institutions.
- Population health analytics: aggregated de-identified data products for FMOH, donor agencies, research.
- Patient app maturity: dependents, multi-language at full coverage (Hausa, Yoruba, Igbo, Pidgin), accessibility features, integrations with national health programs (immunization, maternal health).
- Pharma supply chain integration: traceability from manufacturer to dispense for high-value drugs.

7.2 Exit Criteria

- 1,000+ active providers across 5+ cities
- 1M+ patient accounts
- 3+ payers live with combined 500,000+ lives under contract
- Hospital tier representing 20%+ of revenue
- Path to profitability visible: gross margin trajectory and unit-economics evidence sufficient for Series B

8. Phase 5 — Regional & Platform (Year 3+)

8.1 Themes

- Cross-border: Ghana first (similar identity infrastructure, growing private health sector). Then Kenya. ECOWAS interoperability as a long arc.
- Public APIs: developer platform, third-party app ecosystem with patient-controlled data sharing (think OAuth-for-health).
- Clinical decision support overlays: medication reconciliation, drug interaction at scale, AI-assisted differential diagnosis as an opt-in clinician tool.
- Research data platform: de-identified, ethics-approved, opt-out controlled dataset for population health research.

- Vertical extensions: oncology continuity, maternal-child longitudinal care, chronic disease management programs (diabetes, hypertension).

9. Critical Path & Dependencies

The following items are on the critical path. Slippage in any of them delays the dependent phases.

Item	Phase	Why critical
NIMC API access (production)	Phase 0–1	Without reliable NIN verification, the identity story collapses. Backup: NIMC stub + manual review fallback for beta.
First HMS adapter	Phase 2	Without an adapter, GP clinics that already have an HMS must dual-key. Adapter availability gates GP rollout pace.
NDPA registration & data residency	Phase 0–1	Pre-launch blocker. No public launch without it.
Independent security audit	Phase 1	Required before payer conversations are credible.
First payer contract	Phase 2–3	Defines whether the long-term revenue model is real.
Offline sync engine	Phase 2	Without it, the provider app fails in real Nigerian connectivity conditions.

10. Team Scaling Plan

Milestone	Headcount	Composition emphasis
End of Phase 0	8–10	Founding team + 2 engineers + designer + DPO
End of Phase 1	20–25	Add field ops, clinical ops, support, more engineers
End of Phase 2	45–55	Sales, payer success, multi-city ops, integrations team
End of Phase 3	90–110	Enterprise sales, more cities, data team, security team
End of Phase 4	180–220	National coverage operations

11. Success Metrics by Phase

Phase	Primary metrics
Phase 1	Active pharmacies, dispense events/week, patient claim rate from pharmacy invites, system uptime
Phase 2	Active clinics, encounters/clinician/day, cross-provider continuity rate, time-to-first-record for new providers
Phase 3	Lives under payer contract, payer revenue, multi-city provider count, net revenue retention

Phase	Primary metrics
Phase 4	National provider count, payer revenue mix, hospital revenue, gross margin
Phase 5	Cross-border lives, API ecosystem size, research data products revenue